

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Graphing Inequalities

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

GRAPHING INEQUALITIES · CH₃ · L8

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The Map

This lesson collapses Graphing Inequalities into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Draw the boundary line

TRIGGER *"draw", "plot", "construct", a diagram/grid*

02 Choose solid or dashed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Test one point

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Shade the correct region

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 The required region is

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 19 website questions, 19 checked solutions, 3 local PDFs read.

01 Draw the boundary line

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x \leq 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q10

The shaded region is left

$$x \leq 0$$

It is below the line

$$y \leq -x + 4$$

Finish: $x \leq 0, y \leq -x + 4, y \geq \frac{1}{3}x - 2$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

10 The shaded region in the diagram is bounded by three lines. The equation of one of the lines is given. $y = -\frac{1}{3}x - 2$ Write down the three inequalities that define the shaded region.

YOUR SOLUTION

02 Choose solid or dashed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2 < x \leq 4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q11

Solve inequality

$$2 < x \leq 4$$

The integer values are

3,4

Finish: (a) 3,4. (b) $y \geq x - 3$, $x + y \geq 5$, $y \leq 6$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

11 (a) Write down the integer values of x that satisfy the inequality $-2 < x < 4$ (2) The region R , shown shaded in the diagram, is bounded by three straight lines. Diagram NOT accurately drawn R O y $y = x - 3$ $x + y = 5$ $y = 6$ (b) Write down the three inequalities that define the region R . (2)

YOUR SOLUTION

03 Test one point

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = -3x + 5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2HR Q7

A straight line with gradient

$$y = -3x + 5$$

Use the inequalities

$$x < 6$$

Finish: (a) $y = -3x + 5$. (b) Shade $x < 6$, $y > 2$, $y \leq x + 1$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

7 (a) Write down an equation of the straight line with gradient -3 and which passes through the point with coordinates (0, 5) (2) (b) Show, by shading on the grid, the region defined by all three of the inequalities $x < 6$ $y \geq 2$ $y < x + 1$ Label the region R O 2 4 6 8 10 10 8 6 4 2 x y (3)

YOUR SOLUTION

04 Shade the correct region

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x \leq 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q10

The shaded region is left

$$x \leq 0$$

It is below the line

$$y \leq -x + 4$$

Finish: $x \leq 0, y \leq -x + 4, y \geq \frac{1}{3}x - 2$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

10 The shaded region in the diagram is bounded by three lines. The equation of one of the lines is given. $y = 1.3x - 2$ Write down the three inequalities that define the shaded region.

YOUR SOLUTION

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$9 - 4x > 17$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q8

Read the graph

$$9 - 4x > 17$$

Read the graph

$$y \geq 2$$

Finish: (a) $x < -2$. (b) $y \geq 2, x \leq 6, y \leq x$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

8 (a) Solve $9 - 4x > 17$ (2) O 1 2 3 4 5 6 7 8 1 2 3 4 5 6 7 8 9 10 y x R (b) Write down the three inequalities that represent the shaded region R (3)

YOUR SOLUTION

06 The required region is

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 1, \quad x = 2, \quad x + y = 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q2

Draw the boundary lines

$$y = 1, \quad x = 2, \quad x + y = 7$$

The required region is

$$y \geq 1$$

Finish: Shade $y \geq 1, x \geq 2, x + y \leq 7$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

2 (a) On the grid, draw and label with its equation the straight line with equation (i) $y = 1$ (ii) $x = 2$ (iii) $x + y = 7$ (3) (b) Show, by shading on the grid, the region that satisfies all three of the inequalities $y \geq 1$, $x \geq 2$, $x + y < 7$. Label the region R. (1)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"draw", "plot", "construct", a dia- gram/grid	Draw the boundary line
"work out", "show", "hence", a familiar exam phrase	Choose solid or dashed
"work out", "show", "hence", a familiar exam phrase	Test one point
"work out", "show", "hence", a familiar exam phrase	Shade the correct region
"work out", "show", "hence", a familiar exam phrase	Read the graph
"work out", "show", "hence", a familiar exam phrase	The required region is

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.